

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 5

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1. TITLE

Implement and Use Modules in Node.js

2. AIM

To understand and implement modules in Node.js by creating a custom module and using a built-in module.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of modules in Node.js.
- To create and export a custom module using `module.exports`.
- To import and use a custom module in another file using `require()`.
- To use a built-in Node.js module (`os`) to retrieve system information.

4. THEORY

In Node.js, a module is a reusable block of code whose functionality can be exported from one file and used in another. Node.js uses the CommonJS module system, where `module.exports` is used to expose functionality from a file, and the `require()` function is used to import it elsewhere.

Node.js also provides several built-in modules that can be used without any installation, such as `fs` (file system operations), `os` (operating system information) and `path` (working with file paths).

Organizing code into modules improves reusability, readability and maintainability, especially as an application grows larger.

5. ALGORITHM / PROCEDURE

1. Create a new Node.js project folder.
2. Initialize the project using `npm init`.
3. Create a custom module file, `calculator.js`, containing arithmetic functions.
4. Export the functions from `calculator.js` using `module.exports`.
5. Create a main file, `app.js`, and import the custom module using `require("./calculator")`.
6. Call the imported functions with sample values and log the results.

7. Import the built-in os module and retrieve system information.
8. Run the application using node app.js.
9. Verify the console output.

6. PROGRAM

calculator.js

```
function add(a, b) {
  return a + b;
}

function subtract(a, b) {
  return a - b;
}

function multiply(a, b) {
  return a * b;
}

function divide(a, b) {
  return b !== 0 ? a / b : "Cannot divide by zero";
}

module.exports = { add, subtract, multiply, divide };
```

app.js

```
const calculator = require("./calculator");
const os = require("os");

console.log("Addition:", calculator.add(10, 5));
console.log("Subtraction:", calculator.subtract(10, 5));
console.log("Multiplication:", calculator.multiply(10, 5));
console.log("Division:", calculator.divide(10, 5));

console.log("\nSystem Information:");
console.log("Platform:", os.platform());
console.log("CPU Architecture:", os.arch());
console.log("Total Memory (MB):", (os.totalmem() / (1024 * 1024)).toFixed(2));
console.log("Free Memory (MB):", (os.freemem() / (1024 * 1024)).toFixed(2));
```

7. CODE EXPLANATION

Tag / Code	Explanation
module.exports	Exports functions/objects from a module so they can be used elsewhere
require("./calculator")	Imports the custom calculator module
require("os")	Imports the built-in os module
os.platform()	Returns the operating system platform
os.arch()	Returns the CPU architecture

os.totalmem() / os.freemem()	Return the total and free system memory in bytes
console.log()	Prints output to the terminal

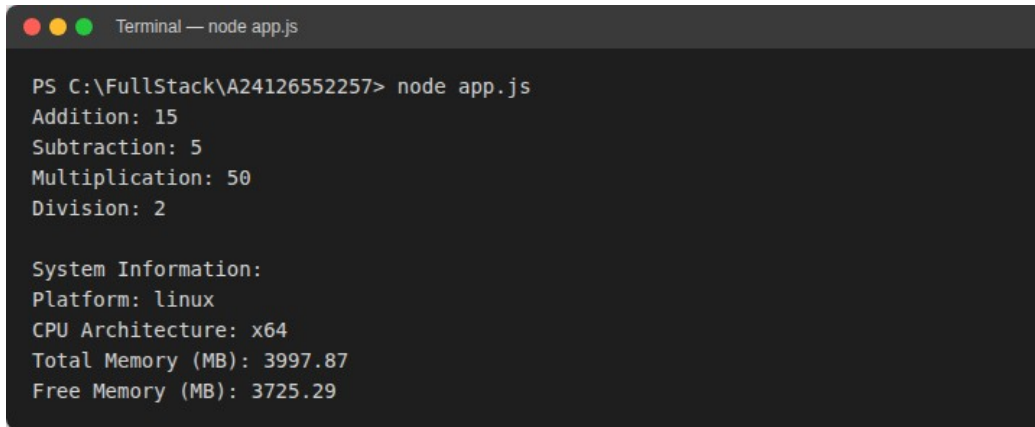
8. TEST CASES AND EXECUTION DATA

The application was run from the terminal using node app.js and the console output was verified.

Test Case	Description	Expected Result	Actual Result	Status
TC-01	Run node app.js	Console shows addition, subtraction, multiplication and division results	All results displayed correctly	Pass
TC-02	Check division by zero	Displays "Cannot divide by zero"	Message displayed correctly	Pass
TC-03	Check os.platform() output	Displays the correct OS platform	Correct platform displayed	Pass
TC-04	Check memory values	Displays valid memory figures in MB	Valid memory figures displayed	Pass
TC-05	Modify calculator.js and re-run	Updated logic is reflected in the output	Updated logic reflected correctly	Pass

Test Case Output

TC-01 — Console shows addition, subtraction, multiplication and division results



```

Terminal — node app.js

PS C:\FullStack\A24126552257> node app.js

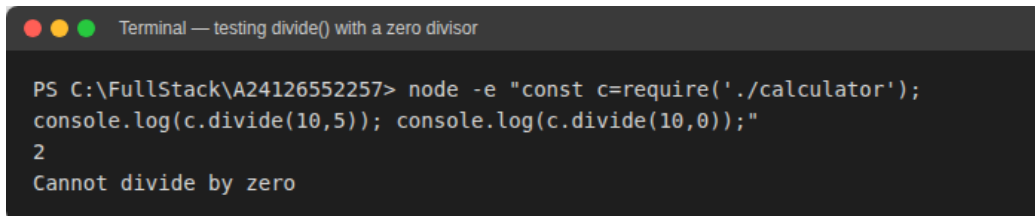
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2

System Information:
Platform: linux
CPU Architecture: x64
Total Memory (MB): 3997.87
Free Memory (MB): 3725.29

```

All four arithmetic results displayed correctly

TC-02 — Displays "Cannot divide by zero"



```

Terminal — testing divide() with a zero divisor

PS C:\FullStack\A24126552257> node -e "const c=require('./calculator');
console.log(c.divide(10,5)); console.log(c.divide(10,0));"
2
Cannot divide by zero

```

calculator.divide(10, 0) correctly returns the guard message instead of throwing an error

TC-03 — Displays the correct OS platform

```
Terminal — node app.js

PS C:\FullStack\A24126552257> node app.js
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2

System Information:
Platform: linux
CPU Architecture: x64
Total Memory (MB): 3997.87
Free Memory (MB): 3725.29
```

Platform: linux correctly reported

TC-04 — *Displays valid memory figures in MB*

```
Terminal — node app.js

PS C:\FullStack\A24126552257> node app.js
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2

System Information:
Platform: linux
CPU Architecture: x64
Total Memory (MB): 3997.87
Free Memory (MB): 3725.29
```

Total and free memory reported in MB

TC-05 — *Updated logic is reflected in the output*

```
Terminal — node app.js (after edit)

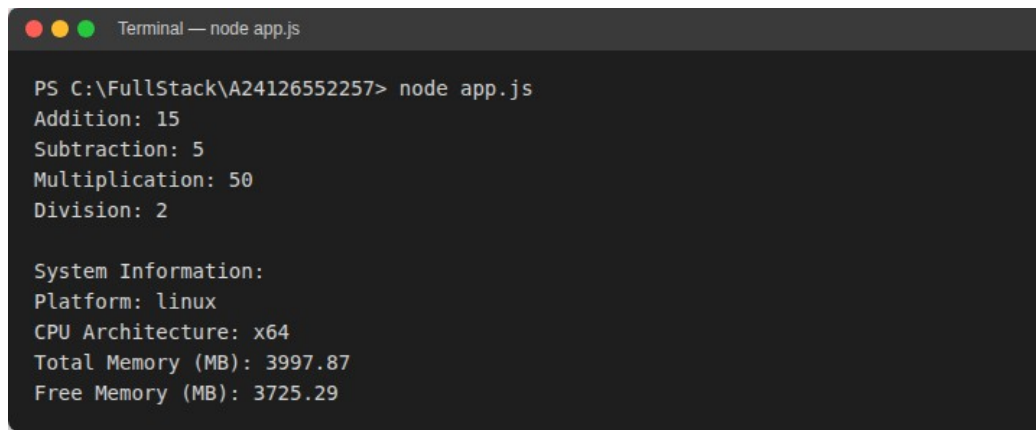
PS C:\FullStack\A24126552257> node app.js
// -- modified app.js to use calculator.add(20, 7) instead of (10, 5) --
Addition: 27
Subtraction: 13
Multiplication: 140
Division: 2.857142857142857

System Information:
Platform: linux
CPU Architecture: x64
Total Memory (MB): 3997.83
Free Memory (MB): 3683.22
```

app.js edited to call calculator.add(20, 7) instead of (10, 5) and re-run — output updated accordingly

9. OUTPUT

Output: Terminal Execution



```
Terminal — node app.js

PS C:\FullStack\A24126552257> node app.js
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2

System Information:
Platform: linux
CPU Architecture: x64
Total Memory (MB): 3997.87
Free Memory (MB): 3725.29
```

Terminal output of node app.js showing calculator results and system information

10. RESULT

Thus, a custom module and a built-in module were successfully implemented and used in Node.js, and the module outputs were verified in the console.